

Subject Description Form

Subject Code	EEE501
Subject Title	Microelectronic Devices and Fabrication
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject aims to let the students gain fundamental knowledge of microelectronic devices and fabrication processes and a deep understanding of the relationship between device design and fabrication processes.
Intended Learning Outcomes <i>(Note 1)</i>	Upon completion of the subject, students will be able to: Category A: Professional/academic knowledge and skills - Understand and describe the semiconductor devices - Understand the details of those important semiconductor fabrication and their roles in device processing - Understand the basic module operation and the fabrication of the important photonic and optoelectronic components Category B: Attributes for all-roundedness - Communicate effectively - Think critically and creatively - Assimilate new technological development in related field
Subject Synopsis/ Indicative Syllabus <i>(Note 2)</i>	Microelectronic device: Introduction of device building blocks, device physics (e.g., MOSFET, Laser, photodetector), and their roles in electronic circuits. Fabrication Process: Different fabrication modules (including crystal growth, photolithography, etching, film deposition, ion implantation, etc) and their roles in device fabrication processing. Technology trends: Technology development towards high-density, high-speed, and low-power consumption.

Teaching/Learning Methodology (Note 3)	Lecture: Through lectures and tutorials, basic principles of microelectronic devices fabrication and processing especially silicon-based platforms will be explained. Concrete examples will be given to illustrate the difference between the different designs of microelectronic devices. Assignment sets will be given to assess the learning progress of students. Assignments and Examinations: Students will work on problem sets in homework, which provides them with opportunities to apply the knowledge gained in lectures.						
	Teaching/Learning Methodology		Intended subject learning outcomes				
			a	b	c	d	e
	Lectures	✓	✓	✓	✓	✓	✓
	Assignments	✓	✓	✓	✓	✓	✓
Assessment Methods in Alignment with Intended Learning Outcomes (Note 4)							
	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				
			a	b	c	d	e
	1. Assignments	40%	✓	✓	✓	✓	✓
	2. Mid-term test	20%	✓	✓	✓		
	3. Examination	40%	✓	✓	✓		
	Total	100 %					
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignments: Let students review the taught materials, do further reading for deeper learning, and understand better of the taught knowledge. Students may find these readings useful and will practice the obtained knowledge in the associated exercises. Test & Exercises: Test & Exercises are designated based on projects to evaluate whether the students are proficient in the taught knowledge to solve practical problems. Mutual discussions are encouraged in order to summarize the findings in a presentation. Examination: Tests will evaluate student’s understanding and knowledge learning from the subject.						

Student Study Effort Expected	Class contact:	
	▪ Lectures	30 Hrs.
	▪ Tutorials and test	9 Hrs.
	Other student study effort:	
	▪ Further reading, doing homework/assignments and preparing for the subject.	81 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	Textbooks • Jaeger, R. C. Introduction to microelectronic fabrication (2nd ed.). Addison-Wesley. 2002. • G.S. May and S.M. Sze. Fundamentals of Semiconductor Fabrication, John Wiley and Sons, 2004. References • Tom Wenckebach, Essentials of Semiconductor Physics, John Wiley and Sons, 1999. • Alamo, J. A. del. Integrated microelectronic devices: physics and modeling. 2018. • Martin-Palma, R. J., & Martínez-Duart, J. Nanotechnology for microelectronics and photonics, Elsevier, 2017.	